

Ideal Gas Law from Statistical Thermodynamics via the 3D Particle-in-a-Box Complete, Step-by-Step Derivation

January 31, 2026

Abstract

We derive $PV = Nk_B T$ from quantum mechanics and statistical thermodynamics without skipped steps. Starting from the 3D particle-in-a-box spectrum we build Z_1 , pass to the continuum with controlled errors, construct Z_N for indistinguishable particles, obtain F , and extract P . A classical phase-space derivation, the Sackur–Tetrode entropy, and validity conditions are included.

1 Quantum Mechanics of a Particle in a Cubic Box

A particle of mass m confined to $V = [0, L]^3$ with infinite walls obeys

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{r}) = E\psi(\mathbf{r}), \quad \psi|_{\partial V} = 0. \quad (1.1)$$

Separation of variables gives normalized eigenfunctions and energies

$$E_{\mathbf{n}} = \frac{\hbar^2\pi^2}{2mL^2} (n_x^2 + n_y^2 + n_z^2), \quad \mathbf{n} = (n_x, n_y, n_z) \in \mathbb{N}^3. \quad (1.2)$$

2 Single-Particle Partition Function: Exact Sum and Classical Limit

The canonical one-particle partition function is

$$Z_1(T, V) = \sum_{n_x=1}^{\infty} \sum_{n_y=1}^{\infty} \sum_{n_z=1}^{\infty} \exp\left[-\beta \frac{\hbar^2\pi^2}{2mL^2} (n_x^2 + n_y^2 + n_z^2)\right], \quad \beta \equiv (k_B T)^{-1}. \quad (2.1)$$

Setting $a \equiv \beta\hbar^2\pi^2/(2mL^2)$, the sum factorizes:

$$Z_1 = \left[\sum_{n=1}^{\infty} e^{-an^2} \right]^3. \quad (2.2)$$

In the classical (high- T /large- L) limit $a \ll 1$, the Riemann-sum bounds

$$\int_0^{\infty} e^{-ax^2} dx - \frac{1}{2}e^{-a} < \sum_{n=1}^{\infty} e^{-an^2} < \int_0^{\infty} e^{-ax^2} dx \quad (a > 0) \quad (2.3)$$

show the relative error vanishes as $a \rightarrow 0^+$. Using

$$\int_0^\infty e^{-ax^2} dx = \frac{1}{2} \sqrt{\frac{\pi}{a}}, \quad (2.4)$$

we obtain

$$Z_1 \simeq \left(\frac{1}{2} \sqrt{\frac{\pi}{a}} \right)^3 = \frac{1}{8} \left(\frac{2mL^2}{\beta \hbar^2 \pi} \right)^{3/2}. \quad (2.5)$$

With $V = L^3$, $\beta^{-1} = k_B T$, and $h = 2\pi \hbar$,

$$\boxed{Z_1 = V \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} = \frac{V}{\lambda_T^3}, \quad \lambda_T \equiv \frac{h}{\sqrt{2\pi m k_B T}}} \quad (2.6)$$

Density of states route. Counting $\mathbf{n}$ -points in the first octant: $N(E) = \frac{1}{8} \cdot \frac{4}{3} \pi R^3$ with $R^2 = (2mL^2/\hbar^2 \pi^2)E$. Then $g(E) = dN/dE = \frac{V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \sqrt{E}$ and $Z_1 = \int_0^\infty g(E) e^{-\beta E} dE$ gives ??.

3 N Noninteracting Indistinguishable Particles

For N noninteracting particles in the Maxwell–Boltzmann regime,

$$\boxed{Z_N(T, V) = \frac{Z_1^N}{N!} = \frac{1}{N!} \left(\frac{V}{\lambda_T^3} \right)^N} \quad (3.1)$$

with degeneracy parameter $n\lambda_T^3 \ll 1$ and $n \equiv N/V$.

4 Helmholtz Free Energy and Pressure

The Helmholtz free energy is

$$F(T, V, N) = -k_B T \ln Z_N = -k_B T [N \ln V - 3N \ln \lambda_T - \ln N!]. \quad (4.1)$$

Using Stirling's expansion

$$\ln N! = N \ln N - N + \frac{1}{2} \ln(2\pi N) + \mathcal{O}\left(\frac{1}{N}\right), \quad (4.2)$$

we can write

$$F(T, V, N) = -Nk_B T \ln\left(\frac{V}{N}\right) + 3Nk_B T \ln \lambda_T + \frac{1}{2} k_B T \ln(2\pi N) + \mathcal{O}\left(\frac{k_B T}{N}\right). \quad (4.3)$$

Since λ_T is independent of V ,

$$\boxed{P = - \left(\frac{\partial F}{\partial V} \right)_{T, N} = \frac{Nk_B T}{V}} \quad (4.4)$$

5 Equation of State

Rearranging ??,

$$\boxed{PV = Nk_{\text{B}}T} \quad (5.1)$$

and with $N = n N_{\text{A}}$, $R \equiv N_{\text{A}}k_{\text{B}}$ one gets $PV = nRT$.

6 Classical Phase–Space Derivation

The canonical integral for one particle is

$$Z_1 = \frac{1}{h^3} \int_V d^3x \int_{\mathbb{R}^3} e^{-\beta \mathbf{p}^2/2m} d^3p = \frac{V}{h^3} \left(\int_{-\infty}^{\infty} e^{-\beta p^2/2m} dp \right)^3. \quad (6.1)$$

Using $\int_{-\infty}^{\infty} e^{-\alpha p^2} dp = \sqrt{\pi/\alpha}$ with $\alpha = \beta/(2m)$ yields $Z_1 = V(2\pi m/\beta h^2)^{3/2} = V/\lambda_{\text{T}}^3$, consistent with the quantum counting.

7 Entropy: Sackur–Tetrode

From ??,

$$S = - \left(\frac{\partial F}{\partial T} \right)_{V,N} = Nk_{\text{B}} \left[\ln \left(\frac{V}{N\lambda_{\text{T}}^3} \right) + \frac{5}{2} \right] + \mathcal{O} \left(\frac{1}{N} \right). \quad (7.1)$$

8 Validity

Assumptions: noninteracting particles; classical (nondegenerate) limit $n\lambda_{\text{T}}^3 \ll 1$; $L \gg \lambda_{\text{T}}$ for replacing sums by integrals; and the thermodynamic limit $N, V \rightarrow \infty$ with N/V fixed so that $\mathcal{O}(1/N)$ terms vanish.